DATA PLAN OPTIMIZATION ANALYSIS PROCESS

David Allen Personal Data Project Files

The aim of this project is to maximize data usage at minimal cost. But this is also dependent on the test subjects data usage pattern. The most important part of data analysis is the thinking pre-planning process. The data analysis plan was carried out as follows:

1. Data Ingestion and cleaning.
2. Structural Break Analysis.
3. Exploratory Data Analysis.
4. Time Series Decomposition and AR(1) modelling
5. Statistical Synthesis.
6. Optimization.

I. **Data Ingestion and Cleaning:** Daily data usage SMS logs were exported to excel. The period used was from November 2025 to December 2025. The first five in the table are shown below :

No	Usage date	Mb
0	2025-11-06	99.35
1	2025-11-08	163.15
2	2025-11-09	215.85
3	2025-11-10	735.52
4	2025-11-11	310.30

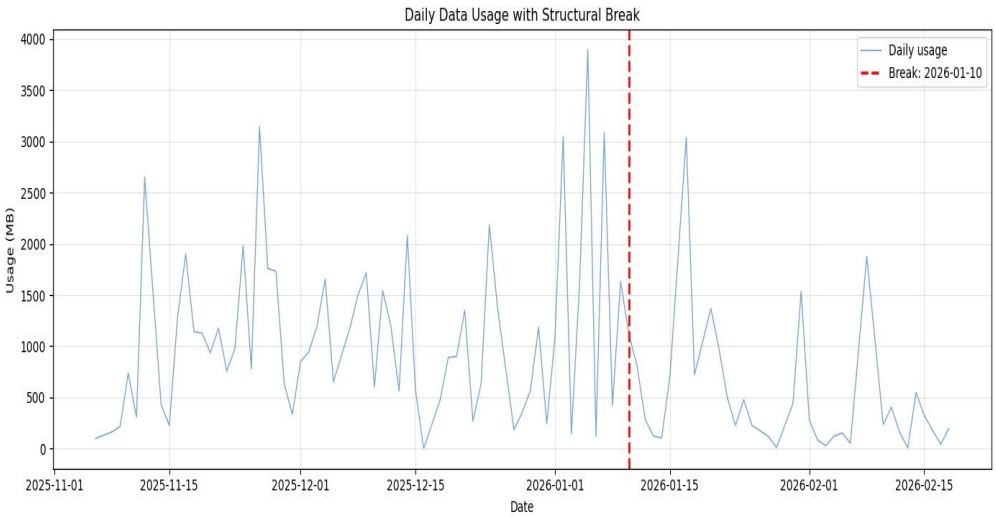
The overall data usage statistics is given below:

Day	Mean	median	Std. Dev	Count
Monday	1149.2	735.5	1118.1	11
Tuesday	583.9	310.3	568.2	15
Wednesday	1146.2	768.9	1147	14
Thursday	835.9	700.9	597.5	15
Friday	957.9	772.4	874.7	12
Saturday	792.1	564.1	788	13
Sunday	907.8	809.8	631.1	15

This summary of the total statistics is somewhat peculiar because it shows that the standard deviation of the distributions are just as large as the means for a lot of days.

II. Structural Break Analysis: The test subject for the data moved to a new job somewhere in December. So the researcher opined that there might be a structural break in the data. The process for the structural break analysis includes－Pre-Post Summary statistics(Median, 90th percentile, variance), Chow test and Cusum test for actual data break. The results are shown below:

Break date: 2026-01-10
Pre- break period: 2025-11-06 to 2026-01-09 (60 days)
Post- break period: 2026-01-10 to 2026-02-18 (33 days)




```
           --- PRE-BREAK ---
Mean:     1106.9 MB
Median:   939.7 MB
Std Dev:  847.9 MB
Min:      1.6 MB
Max:      3893.8 MB
90th %ile:2090.4 MB

--- POST-BREAK ---
Mean:     524.5 MB
Median:   279.5 MB
Std Dev:  649.2 MB
Min:      10.1 MB
Max:      3038.3 MB
90th %ile:1318.5 MB

======================================================================
STATISTICAL TESTS
======================================================================

Welch's t-test (difference in means):
  t-statistic = 3.702, p-value = 0.0004
  → Significant at α = 0.05

Mann-Whitney U test (difference in medians):
  U-statistic = 1492, p-value = 0.0001
  → Significant

Levene's test (difference in variances):
  W-statistic = 3.644, p-value = 0.0594
  → Not significant

Chow test (F-test for structural break at 2026-01-10):
  F-statistic = 11.755, p-value = 0.0009
```

→ Break significant

This is the time series of the data usage of the test subject. The data shows a constant awkward behaviour and pattern if you zoom in closely which breaks in December . December seems to be the transmission phase. From the break in the first week of January, we can see a new pattern where it goes up then slides down constantly with little bumps up, then rises again. The second major spike after the break also mirrors the same pattern but in a smaller degree. And the third major spike mirrors the same thing also. Almost like a pattern, but we will formally investigate that soon. From our summary statistics, there are wide fluctuations but a consistent pattern. It might be a seasonal random walk or a seasonal log-normal distribution.

So we look at the tests and see that the difference in means is significant, and the difference in medians is significant. But the difference in variances is not significant. This means that means and median usages of data has changed but the pattern of usage is the same. This is the same kind of thing we are saying, the distribution has not changed but the means and medians have changed. So it is still a log-normal distribution with maybe different means in different weeks, different medians in different weeks.

So the question is what kind of distribution has a stable pattern Monday to Sunday, like a curve, but has a lognormal distribution Mondays to mondays. So I mean week after week have the same pattern constant. But the distribution of data usage on each day is also a stable lognormal distribution.

III. Exploratory Data Analysis: This part of the analysis visualizes the distributions, day of week patterns. We have included histograms, box plots, and time series plots.

-- PRE- BREAK ---

Mean: 1106.9 MB

Median: 939.7 MB

Std Dev: 847.9 MB

Min: 1.6 MB

Max: 3893.8 MB

90th %ile:2090.4 MB

--- POST- BREAK ---

Mean: 524.5 MB

Median: 279.5 MB

Std Dev: 649.2 MB

Min: 10.1 MB

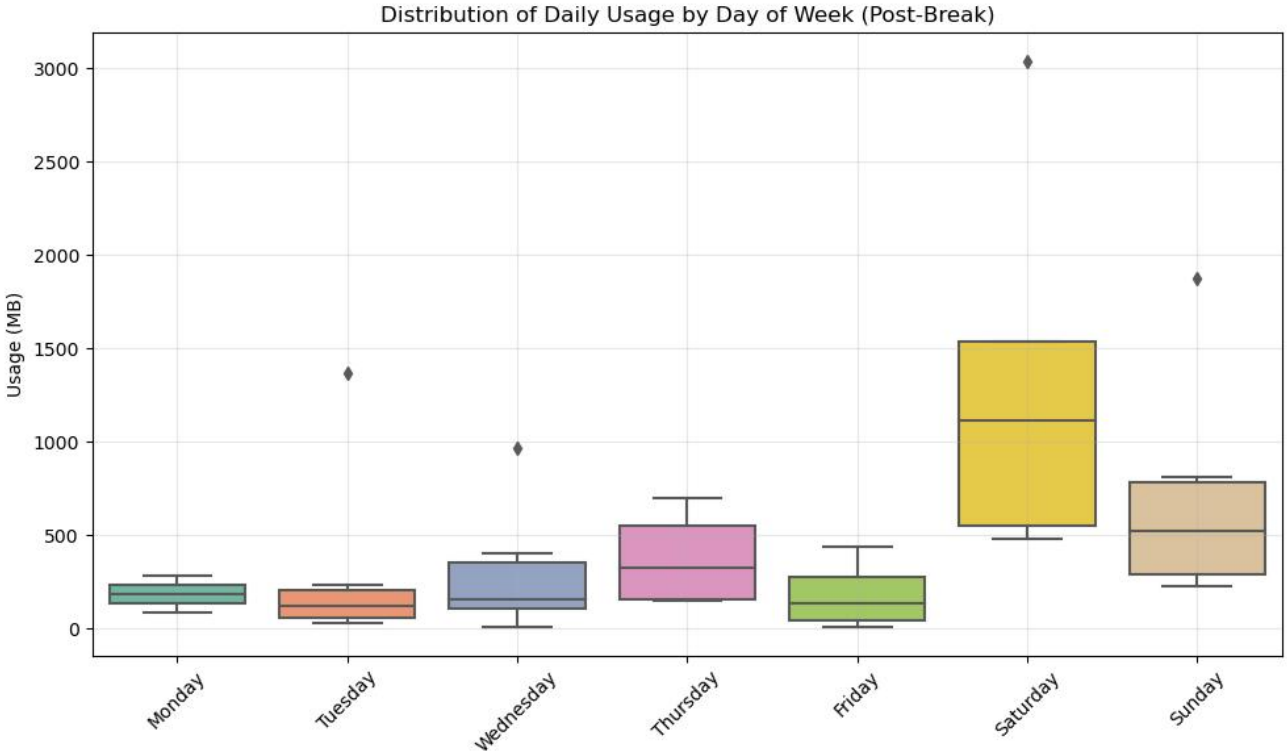
Max: 3038.3 MB

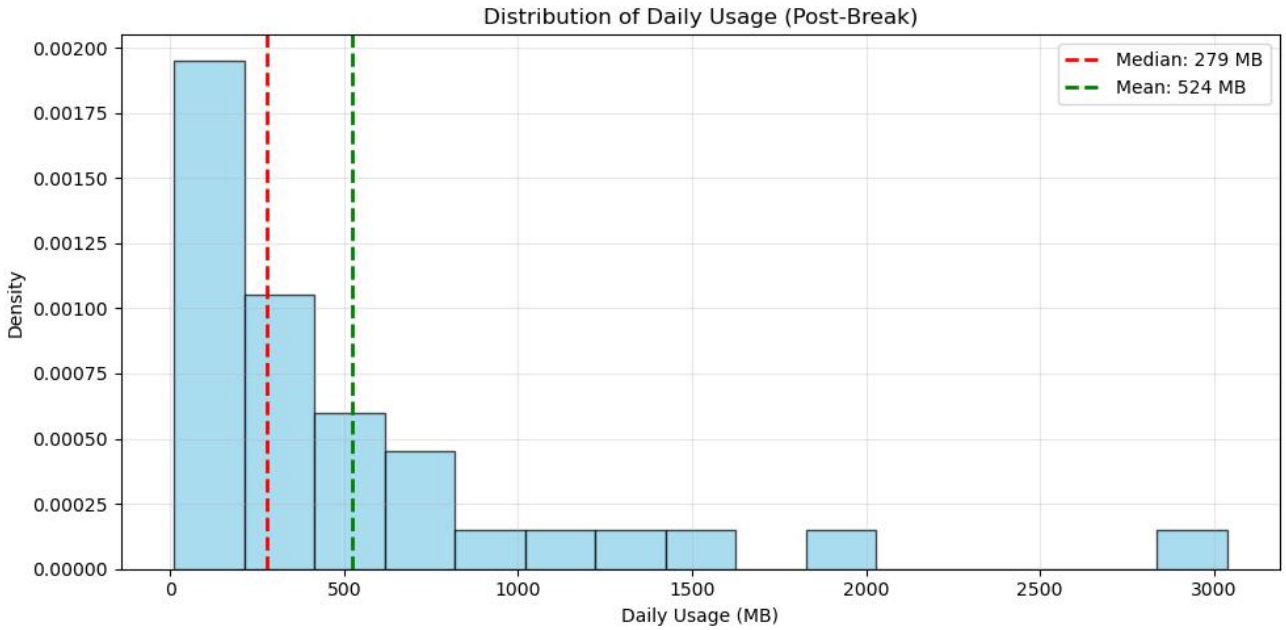
90th %ile:1318.5 MB

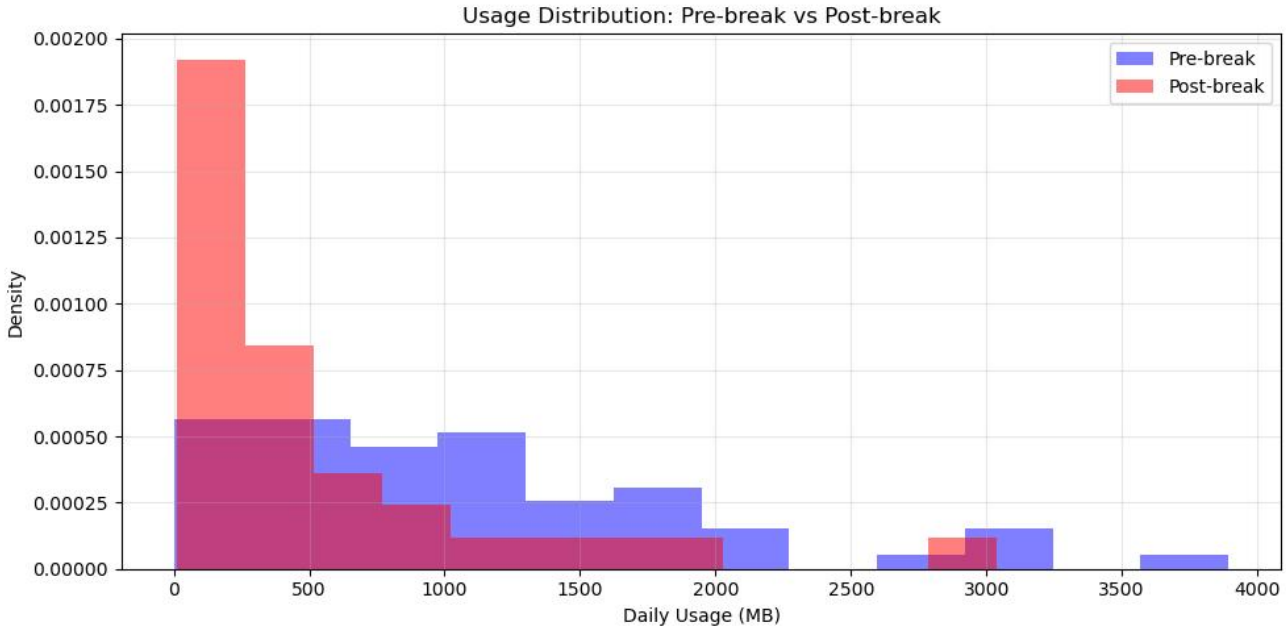
DAY OF WEEK STATISTICS(POST-BREAK)

Day	Median	Mean	Std. Dev	p90
Monday	184.5	184.5	141.2	264.3
Tuesday	121.7	320.2	518.9	802.6
Wednesday	160	302.2	353.6	688.2
Thursday	329.7	378.7	268.8	640.6
Friday	139.9	182.6	195.7	376.3
Saturday	1117.9	1344.1	1041.6	2437.6
Sunday	521.8	706.5	622.4	1343.2

Post break data shows that the top 90% percentile is far exceeding the median.


Distribution of Daily Usage by Day of Week (Post-Break)
Usage (MB)
3000
2500
2000
1500
1000
500
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

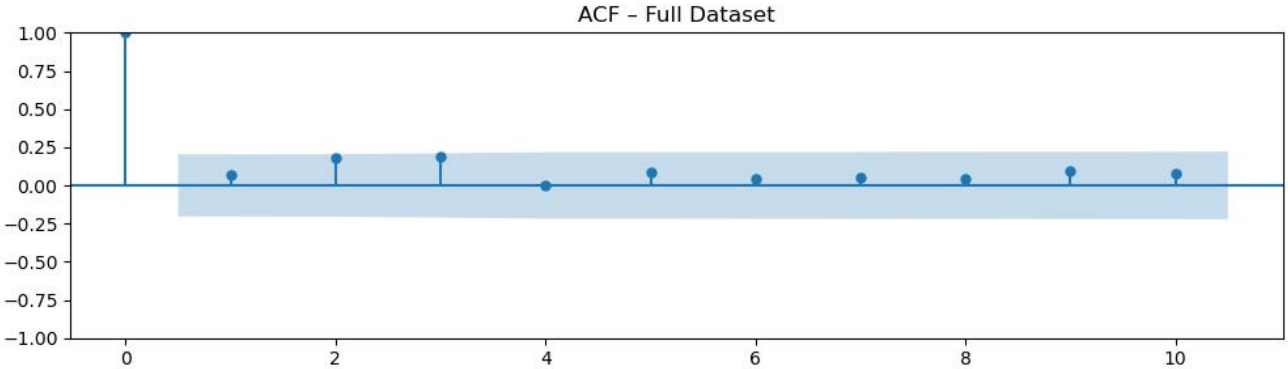

Distribution of Daily Usage (Post-Break)
Median: 279 MB
Mean: 524 MB
Density
0.00200
0.00175
0.00150
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
0
500
1000
1500
2000
2500
3000
Daily Usage (MB)

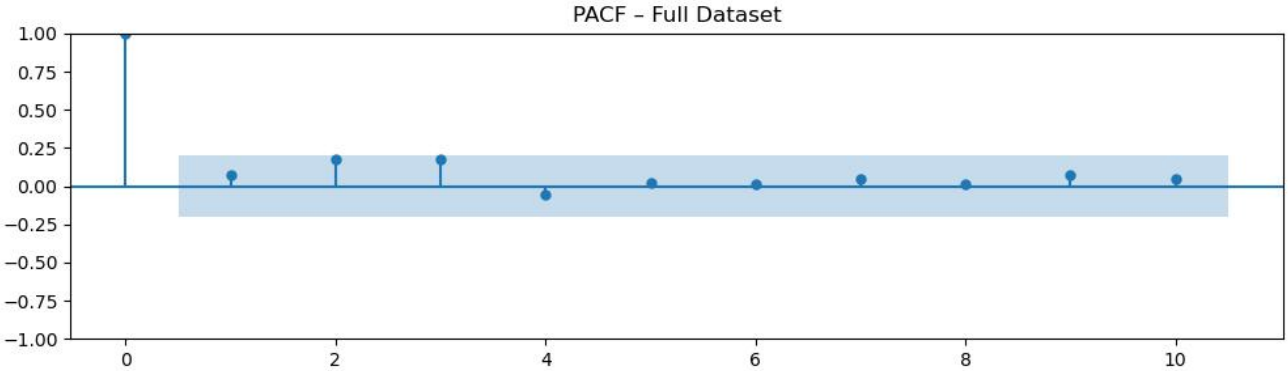


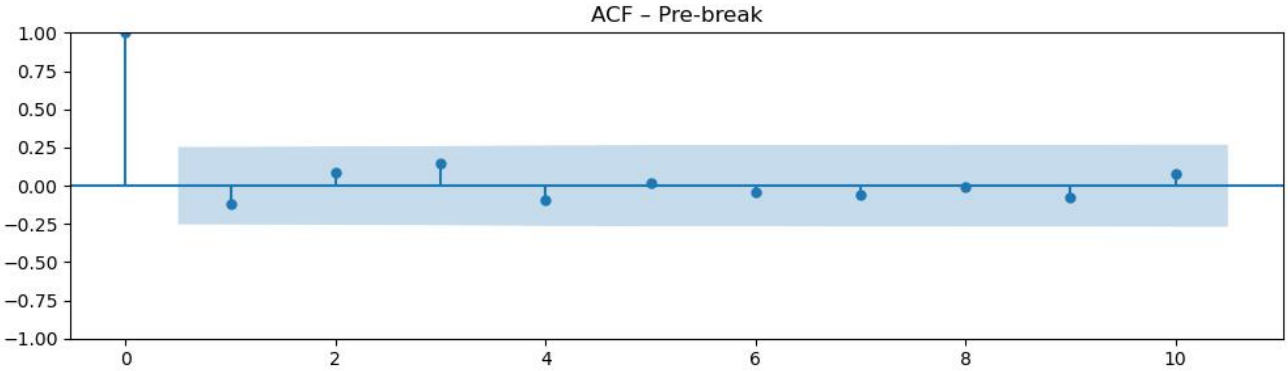
My post-break data is looking like a log-normal distribution. A very clear log-normal distribution. The pre-data break had a log normal distribution too, where very high level uses of data where Extreme data points where very rare. It is perhaps because of the transition phase. If the user's behavior is consistent, then even changing jobs might not affect it that much.

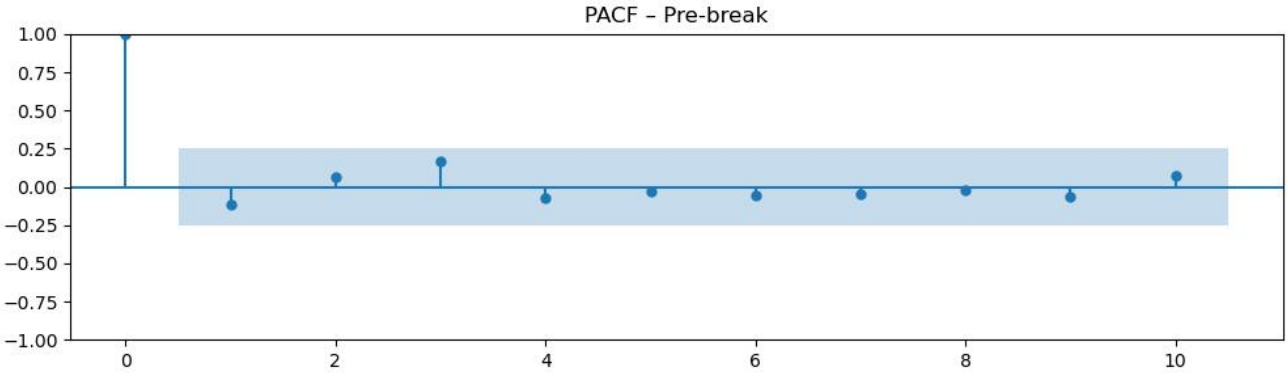
IV. Time Series Decomposition : The ACF and PACF functions are shown below:

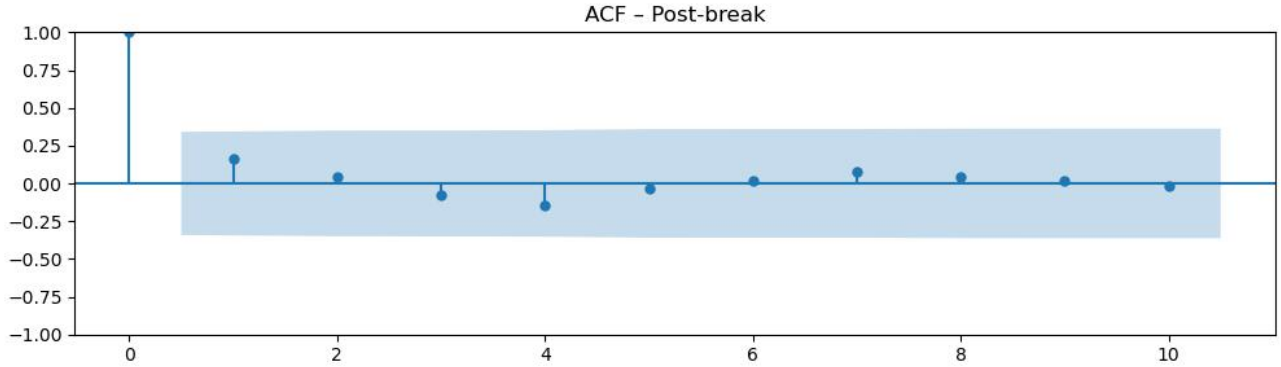
```
TIME SERIES DECOMPOSITION - DATA USAGE
======================================================================
Full data: 93 days (2025-11-06 to 2026-02-18)
Pre-break: 60 days (2025-11-06 to 2026-01-09)
Post-break: 33 days (2026-01-10 to 2026-02-18)
```

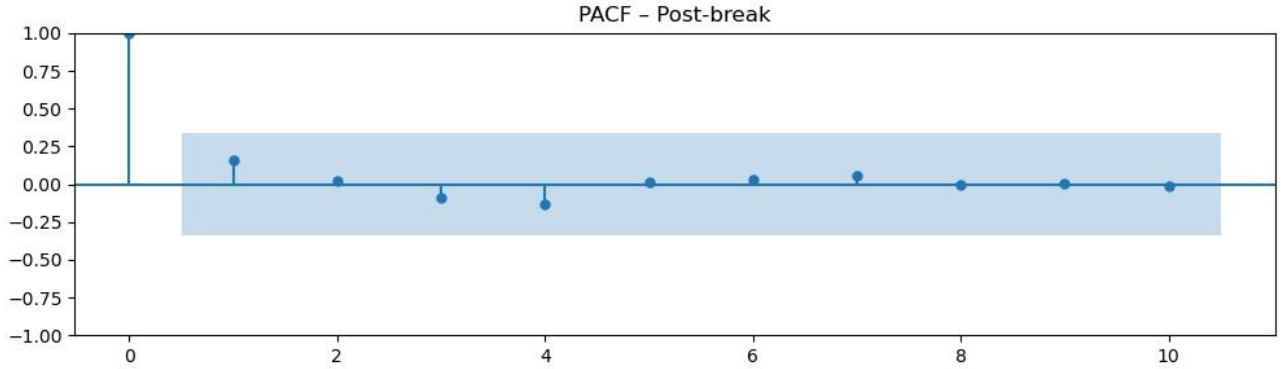


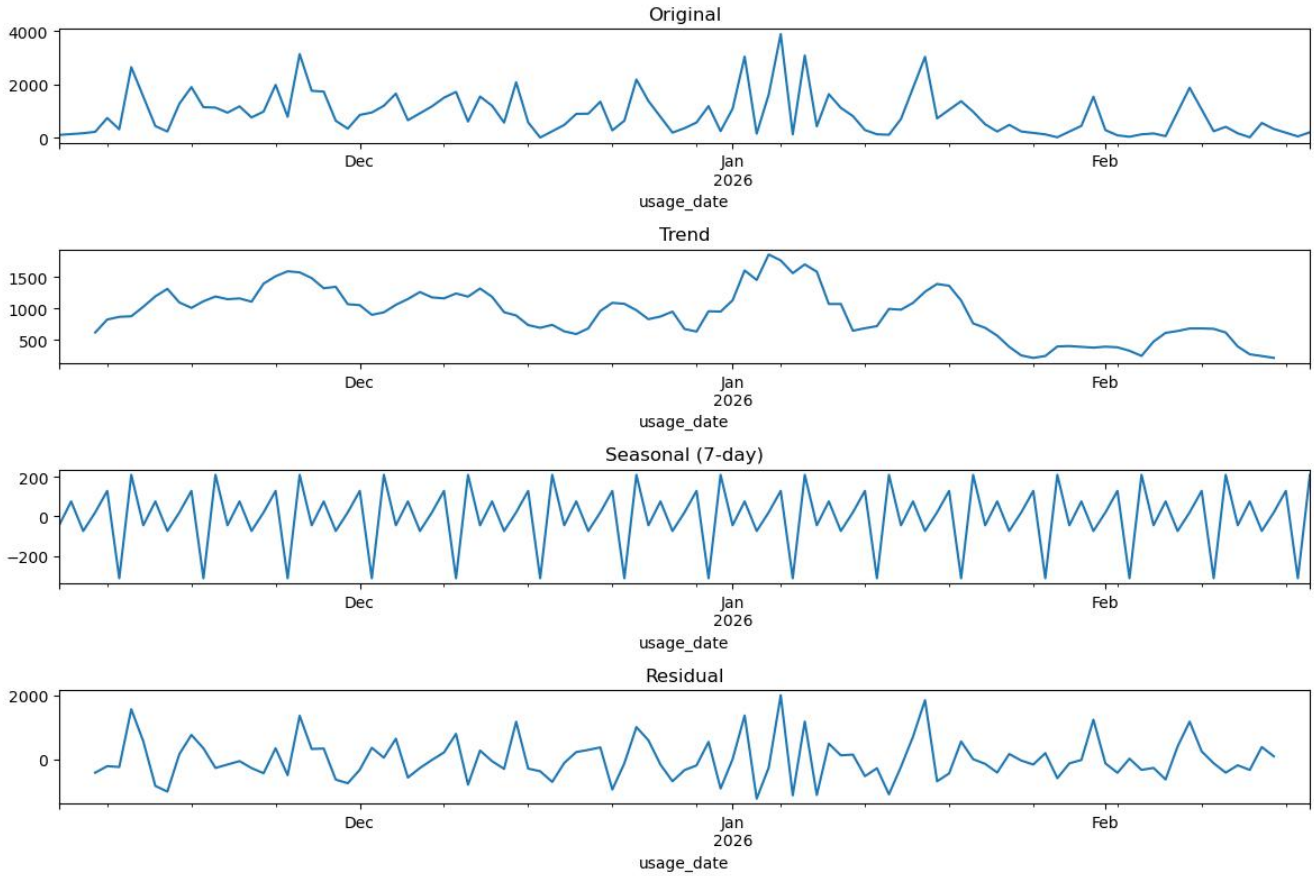







ACF – Post-break
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10


PACF – Post-break
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10


Original
4000
2000
0
Dec
Jan
2026
Feb
usage_date
Trend
1500
1000
500
Dec
Jan
2026
Feb
usage_date
Seasonal (7-day)
200
0
−200
Dec
Jan
2026
Feb
usage_date
Residual
2000
0
Dec
Jan
2026
Feb
usage_date

Seasonal decomposition completed – note the strong weekly pattern.

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INTERPRETATION

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• ACF plots show a clear spike at lag 7 in all regimes, confirming weekly seasonality.
• No significant autocorrelation at low lags (especially lag 1) indicates day-to-day independence.
• AR(1) coefficients are all near zero, supporting the view that each day's usage is essentially an independent draw from its day-of-week distribution.
• Pre-break $\varphi = -0.120$, post-break $\varphi = 0.161$ – similar, suggesting the dependence structure did not change after the break.

These results justify the use of bootstrap resampling (treating days as independent) for simulating monthly totals in the next step (Statistical Synthesis).

V. **Statistical Synthesis:** We did four things in this section. We compared the pre and post-break medians and 90th percentiles to confirm day of week patterns across regimes. We then did official tests via Q-Q plots to confirm daily data usage followed a log-normal distribution. We also confirmed that boostrapping the data monthly revealed the monthly data usage followed a normal distribution when simulated for 10,000 months based on sums of random daily data usage. The analyses are shown below:

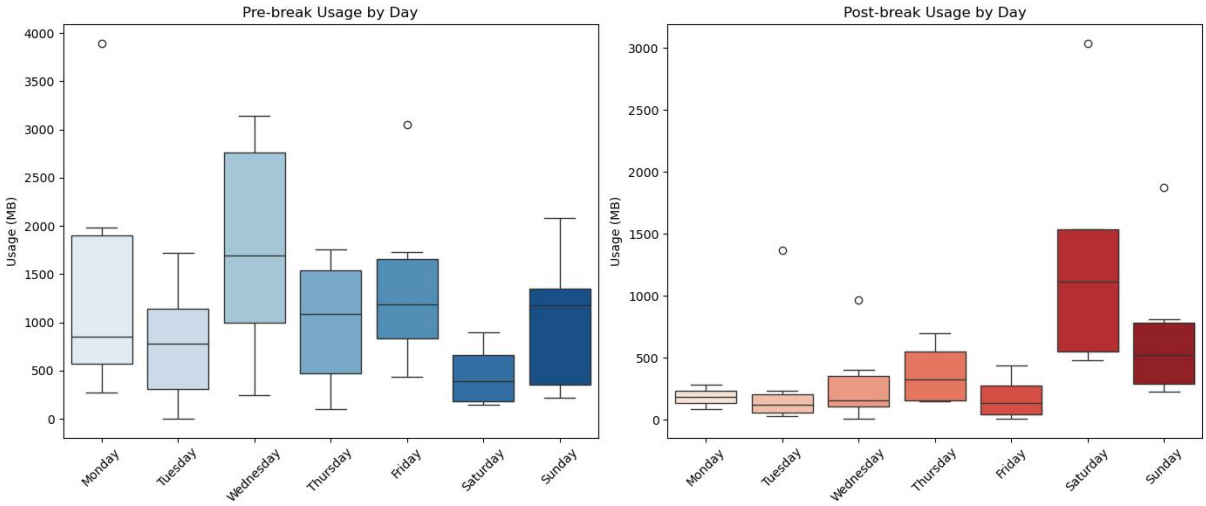
The post break medians show and tell a different story than that of pre-break and they are obviously smaller except for Saturday. Now this is probably because since the user's new job is tasking and he does not really have time for personal data usage. The user's new job has constant wifi which explains the lesser data usage post-break.

```
=====================================================================
Stability of Day-of-Week Pattern Across Regimes
======================================================================
Pre-break: 60 days (2025-11-06 to 2026-01-09)
Post-break: 33 days (2026-01-10 to 2026-02-18)

Day-of-week medians and 90th percentiles (MB):
           median_pre  <lambda_0>  median_post  <lambda_0>
weekday
Monday          851.8      2365.3        184.5       264.3
Tuesday         780.1      1290.8        121.7       802.6
Wednesday      1691.3      3104.5        160.0       688.2
Thursday       1087.8      1674.9        329.7       640.6
Friday         1187.0      2126.0        139.9       376.3
Saturday        395.5       799.6       1117.9      2437.6
Sunday         1173.7      1708.6        521.8      1343.2

======================================================================
Mann-Whitney U tests (p-value < 0.05 indicates distribution change)
======================================================================
           p-value  n_pre  n_post
weekday
Monday      0.0727      9       2
Tuesday     0.1810      9       6
Wednesday   0.0047      8       6
Thursday    0.1483      9       4
Friday      0.0081      8       4
```

Saturday	0.0932	8	5
Sunday	0.2721	9	6



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Interpretation

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Significant differences (p < 0.05) found for: Wednesday, Friday

Therefore we will keep pre- and post-break periods separate for distributi
onal testing.

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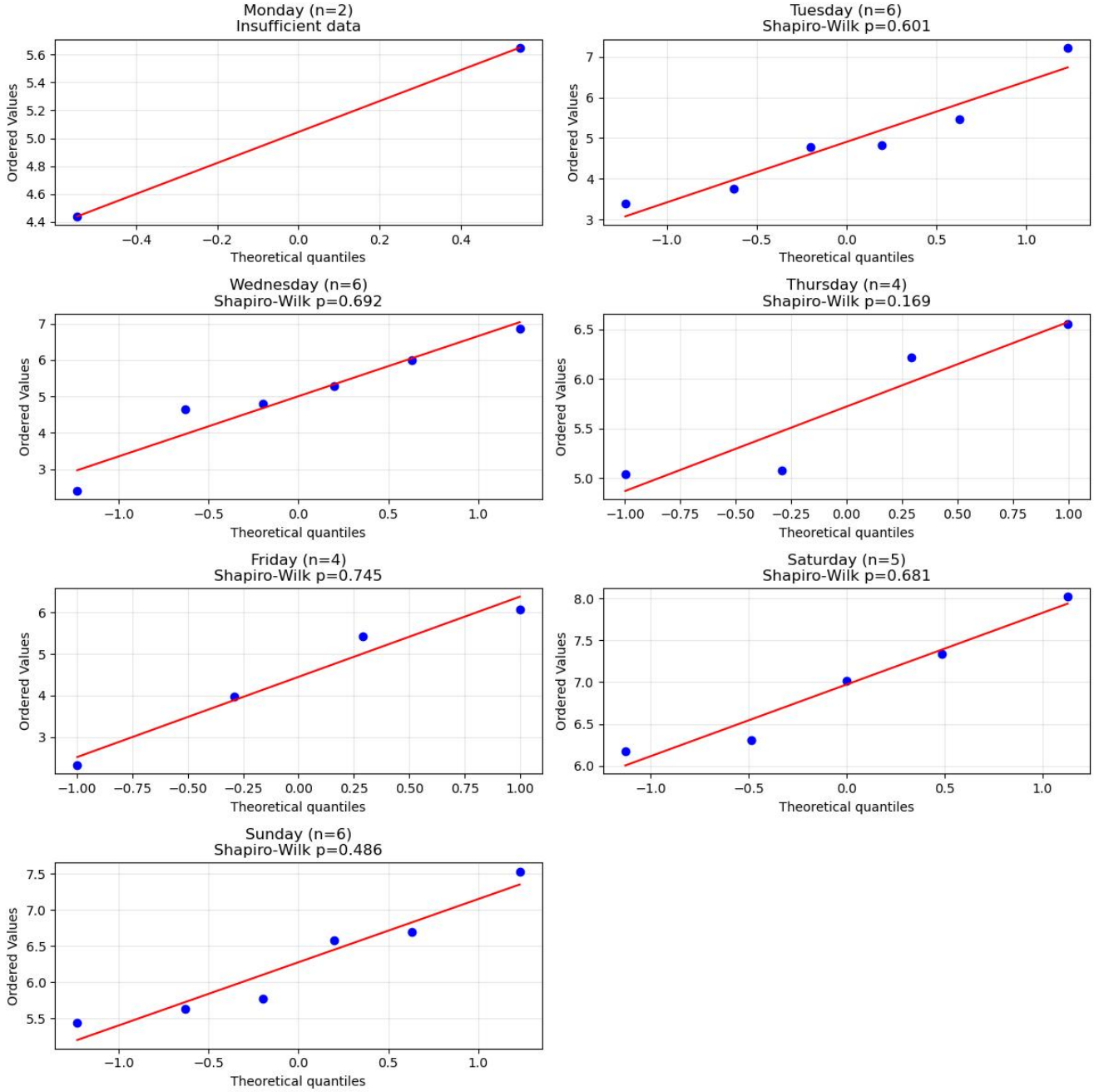
======================================================================
Testing Lognormal Assumption per Weekday (Post-Break Only)
======================================================================
Post-break period: 2026-01-10 to 2026-02-18
Total days: 33
```



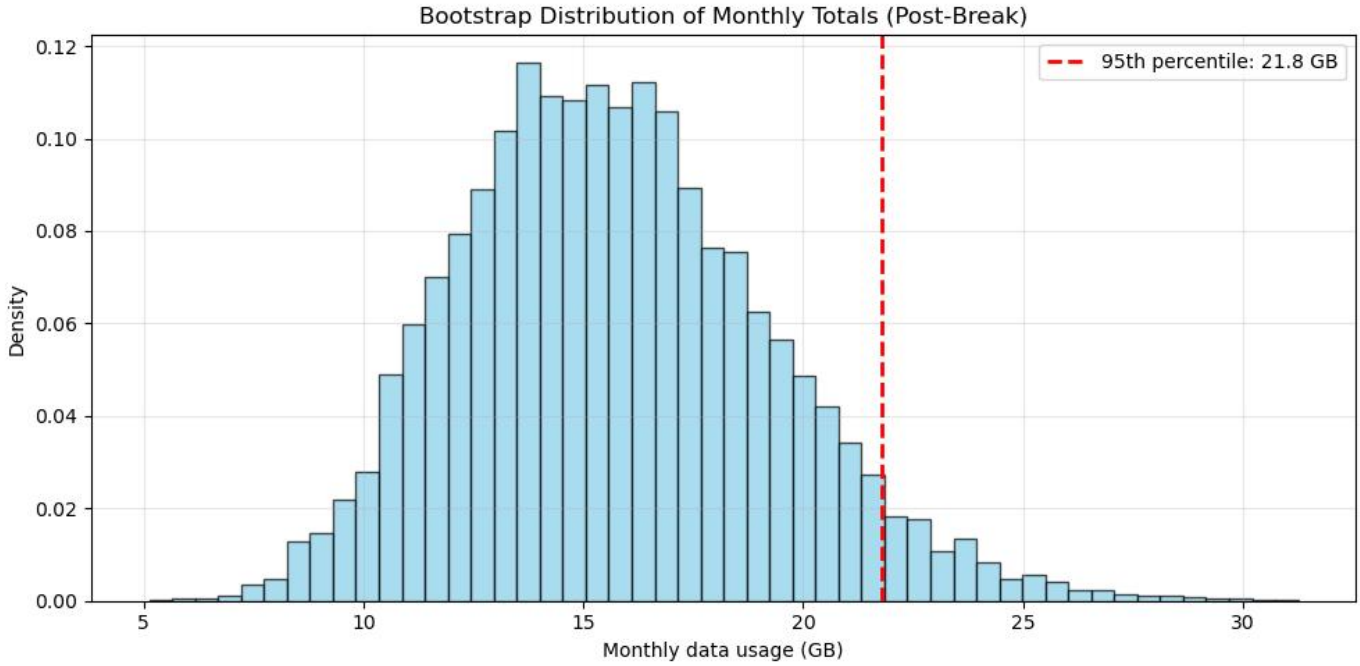
```
Shapiro-Wilk Test Results (for log-transformed usage):
           n  statistic  p-value
weekday
Monday     2        NaN      NaN
Tuesday    6     0.9326   0.6006
Wednesday  6     0.9441   0.6920
Thursday   4     0.8303   0.1686
Friday     4     0.9547   0.7453
Saturday   5     0.9422   0.6813
Sunday     6     0.9173   0.4864



======================================================================
Interpretation
======================================================================
The Q-Q plots show how closely the log-transformed usage follows a normal
distribution.
Points that lie close to the diagonal support the lognormal assumption.
The Shapiro-Wilk p-values are reported but have low power due to small sam
ple sizes.
If the plots appear reasonably linear, we can consider the lognormal assu
mption as a plausible description.
```

```
Post-break period: 2026-01-10 to 2026-02-18
Number of days: 33

======================================================================
Bootstrap Results (10,000 simulated months)
======================================================================
Mean monthly need: 15.69 GB
Median monthly need: 15.46 GB
Std deviation: 3.52 GB
95th percentile: 21.79 GB ← target for conservative planning
```



VI. Optimization: Our goal of this whole analysis is to minimize cost per mb and minimise waste of resources too. We created 10,000 bootstrapped samples of (U_i) from the post-break dataset. Our variable is the simulated usage (U_i) per each iteration. The Constants are V_{base} and P_{base} . Base referring to the base data bundle. For every iteration (i) where usage is compared against a base plan, the following metrics must be recorded:

I. Unit Cost per Mb: $P_{base} / V_{base.}$

II. Effective Unit Cost : $P_{base} / U_{i.}$

III. Total Waste (W): $\max(0, V_{base} - U_i)$.

IV. ShortFall (S): $\max(0, U_i - V_{base})$.

V. Recovery Plan cost (P_{top}): The cost of the cheapest available plan, where Volume ≥ S.

The Decision Metric is the Blended Effective cost. For the goal of minimising cost, we decided to use cost per mb. We went further by saying because we want to minimize waste as well, we decided to introduce effective Unit cost. This measure excludes wasted data and penalises waste, the higher the wasted data, and smaller the usage, the higher the effective unit cost. Now this metric still had its shortcomings. For plans that did well on this but failed everytime, they would have to restock or buy another plan. We took this into account and formulated that we calculate the shortfall, and we then shared it across the 10,000 U bootstrappings. We took the months where U exceeded V. Calculated the shortfall S, divided S by 10,000, to get the min monthly shortfall risk on average and added it to the cost.

P_{top} is the cost of the cheapest plan that can reliably cover S. But we do not select this manually. We created an algorithm to do the selection of the plan based on what day or remaining days the month had before the plan exhausted, and how much was left. And the remaining days and how much determined P_{top} . So we needed to add the cost per Mb of this top-up to the effective cost per Mb to get the true effective cost per Mb. This was now done by simulation.

To test this, we select a data plan bundle V, and then bootstrap the Post-break data. for every U_i bootstrap from our data, we found out if U>V, if yes, for that particular V, we found the cheapest top up price that could reliably cover it as P_{top}^i . We did that for all bootstrapped monthly totals of usage from 1 to 10,000. Where U=V or U<V, we did nothing, as the BEC will eventually penalise wastage since we are recording only price and usage not data bundle volume. At the end of 10,000 bootstraps, we added the total of base

prices plus top up costs, then divided it by the total usage. This gave us a simulation-tested effective cost per mb accounting for both waste and failure. And using the bootstrap rather than using aggregated means to find top up was better. We did not just find the average S as we planned to earlier, but we found S every time and the cheapest top-up in every scenario and as such, the true probable effective cost estimate from the distribution based on my data usage was discovered. We tried this with MTN data bundles and the algorithim yielded the data bundle with the best plan.

Ideally the BEC formula is = $$BEC = \frac{\sum(P_{base} + P_{top})}{\sum U}$$

Logic: P_{top} = 0 for all successful months (U \le V). For failure months (U > V), P_{top} is the cost of the cheapest plan that totally offsets the shortfall.

Where P_{base} is the price of the data bundle we are analyzing. P_{top} is the price of the top-up data bundle to reliably cover S. So we simulated the exact BEC from the bootstrap. We had the following results:

BLENDED EFFECTIVE COST (BEC) OPTIMISATION

Plan	Volume (GB)	Price (₦)	BEC (₦/GB)	Total Cost (₦)	Total Usage (GB)
Monthly 14.5GB	14.5	5000	419.91	65903000	156947.39
14.5GB + 1×7GB	21.5	6800	441.57	69303700	156947.39
4×7GB (no base)	28.0	7200	458.91	72025400	156947.39
Monthly 24GB	24.0	7500	480.18	75363700	156947.39
14.5GB + 2×7GB	28.5	8600	548.03	86011700	156947.39
Monthly 65GB	65.0	16000	1019.45	160000000	156947.39
Monthly 165GB	165.0	35000	2230.05	350000000	156947.39

```
==============================================================================
RECOMMENDATION
==============================================================================
The plan with the lowest Blended Effective Cost is:
 → Monthly 14.5GB
 Volume: 14.5 GB
 Price: ₦5,000
 BEC: ₦419.91 per used GB
 Total cost over 10000 months: ₦65,903,000
 Total usage over 10000 months: 156947 GB
```

The best plan to choose is the 14.5gb plan for 5,000 naira in this subjects case.

SUMMARY

In summary, we aimed to find out using data usage stats, the best data plan for a user. It seems intelligent to just sum stuff up and add, but sometimes even monthly data added might not suffice, and varying data usages across days and weeks might affect it. In this user's case, he

obtained a new job with office wifi, so there was a change in how he used data. So we documented this break in pattern of data. We also noticed that as random as data consumption seems, since it is determined by daily routine. A constant walk using phone to your junction or on your way to walk, might affect your data usage. Putting off the data on Sunday while in church might reduce it. Watching movies on a particular weekday might effect it. Yes data usage might seem random at first sight but for most sane rational people, there is a pattern which would differ from person to person.

We also saw that using a bootstrap and few data points. We could bootstrap randomly 10,000 times to simulate the true distribution and pattern of data usage which we did. We then used the bootstrap to do the analysis and calculations. For each bootstrapped month, we simulated the best data choice to make, and found out the best plans that did well on average over 10,000 months with different U scenarios. Hence we arrived at our conclusion. Hope you enjoyed it lol. DM with 5k for personalized data recommendation access lol. My own is a 5k 14.5Gb MTN monthly data plan. This method works no matter how scattered your data buying can be, cause mine was scattered this year as well, but from now on. Good thing is data is always reliable because no one monitors their data usage that much, especially for people who buy large plans and have a consistent lifestyle. So with this framework people could actually monitor the actual amount of data they need over a given period of time.